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AI-Based Agricultural Marketplace for Farmer–Buyer Matching and Net Realization Optimization

ABSTRACT

The proposed system is an **AI-based agricultural marketplace** developed to connect farmers with suitable buyers and support intelligent and profitable selling decisions. The system collects important information from farmers, including crop type, available quantity, location, crop quality, and selling period. Similarly, buyers can provide their required crop, quantity, location, expected price, quality requirements, and purchase period. The collected information is stored and processed along with historical and current agricultural market data.

Machine learning techniques are used to **predict crop prices and demand** based on relevant market conditions, historical trends, crop information, and other available data. The system then identifies and matches suitable buyers or markets according to crop compatibility, quantity requirements, location, demand, and availability. Geographical distance and transportation requirements are also considered to estimate the cost involved in selling the produce. Based on the predicted selling value and associated costs, the system calculates the **estimated net realization** for each potential selling opportunity.

The system further performs risk analysis and ranks the available buyers or markets based on factors such as predicted price, demand, buyer compatibility, transportation cost, distance, risk, and expected net realization. Finally, it provides the farmer with an explainable recommendation indicating the **most suitable buyer or market**, expected price, estimated net realization, transportation cost, and risk level. The recommendations can be updated when new market prices, buyer requirements, demand, or logistics information becomes available. Thus, the proposed system provides an integrated platform combining **AI-based market intelligence, price and demand prediction, farmer–buyer matching, and net realization optimization** to support data-driven agricultural selling decisions.

Keywords: Artificial Intelligence, Agricultural Marketplace, Farmer–Buyer Matching, Price Prediction, Demand Forecasting, Market Intelligence, Net Realization, Transportation Cost Optimization, Risk Assessment, Machine Learning, AgriTech.

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